NICOLAS RIVEIRO

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EDUCATION

EMBRY-RIDDLE AERONAUTICAL UNIVERSITY Daytona Beach, FL

Bachelor of Science in **Aerospace Engineering** Expected Graduation: **December 2027**

Cumulative **GPA: 4.0**/4.0 **Dean's List**: Fall 2023 to Spring 2026

PROFESSIONAL EXPERIENCE

ROCKET FACTORY AUGSBURG (RFA) - ***Test Engineer Intern*** May 2026 – Aug. 2026

- Designed (P&ID, 3D CAD), manufactured, and operated 7 testing rigs for multiple components (e.g separation locks, hold-down clamps, COPVs, cryogenic valves actuation, cable/press tunnels).
- Led the pad operations for multiple burst/hydrostatic pressure testing on valves/COPVs/tanks.
- Performed cryogenic and vacuum testing to validate sealing and actuation performance under flight conditions.
- Ran vibration and thermal testing (Thermotron) to characterize valve performance and reliability across operational cycles.

ROCKET FACTORY AUGSBURG (RFA) - ***Systems Engineering Intern*** May 2025 – Aug. 2025

- Worked on the design/development of the Flight Termination System (FTS) fluid system for an orbital launch vehicle.
- Integrated the different FTS subsystems into the propellants' fluid system while optimizing its control logic.
- Elaborated its requirements, Fault Tree Analysis, Failure Modes and Effects Analysis, Concept of Operations, Test Cases.
- Followed NASA STD/HDBK, RCC, and company standards to ensure compliant system development and testing.
- Created an interactive P&ID program in order to simulate and visualize possible architecture configurations.
- Completed the FTS licensing documentation package for the UK Civil Aviation Authority (CAA).

DELTA AIR LINES, TECH-OPS - ***Propulsion Repair Engineering (Rolls-Royce) Intern*** Aug. 2025 – Dec. 2025

- Analyzed damages found during inspections and their root cause, supported technicians with instructions.
- Requested repair authorizations and new repair methods to Rolls-Royce for damage not covered in their manuals.
- Designed a more effective and efficient cleaning process for a Stator Vane, included in the Rolls-Royce T-1000 manual.
- Drafted a repair method for P&W GTF engines duct vent, with a potential saving of $4.5M in 4 years on scrapped parts.

PROJECTS

EXPERIMENTAL ROCKET PROPULSION LAB - ***Vertical Test Stand Thrust Structure*** Aug. 2026 – Present

- Led the 3D CAD (Inventor), FEA (ANSYS) and manufacturing of a vertical test stand for a 2,200 lbf vehicle engine.
- Wrote a Python hand-calc tool for cable tensions, stress, buckling and deflection to verify FEA results.
- Coded a plume and heat-transfer model (NASA CEA, finite dif.) to find the most optimal flame diverter configuration.

EXPERIMENTAL ROCKET PROPULSION LAB - ***Test Operator*** Jan. 2025 – Present

- Operated a high-pressure LOX/IPA bi-propellant GSE and feed system across pressurizing, water flows and hotfire tests.
- Executed pre-test checkouts on press+feed systems, instruments and valves, verifying safety-critical parameters/interlocks.
- Monitored real-time sensor data during test in order to catch and flag system anomalies.
- Conducted post-test inspections and data review to characterize the system's performance or identify system anomalies.

EXPERIMENTAL ROCKET PROPULSION LAB - ***Engine Feed System*** Aug. 2023 – Present

- Worked on the assembly and maintenance of a bi-liquid cryogenic propellant feed system and component test rigs.
- Analyzed data to optimize feed system designs/sequences and characterize the engines' and components' performance.

EXPERIMENTAL ROCKET PROPULSION LAB - ***COPV Pressurizing Profile*** Sep. 2026

- Developed a transient model (Python, CoolProp) of GN2 COPV fills, predicting pressure and temperature of the system.
- Modeled transient conduction through a multilayer composite overwrap and choked/unchoked orifice flow.
- Implemented valve control logic with 3 temperature-limit strategies to keep the COPV below its limit.
- Identified source-bottle cascading as the most time-effective way to shorten heat-transfer-limited fills.

ROCKET FACTORY AUGSBURG (RFA) - ***Vehicle and Hold-Down Clamps Separation Locks*** May 2026 – Aug. 2026

- Re-designed a separation lock test rig (pneumatic, structural and DAQ system), decreasing the set-up time by ~70%, Helium usage by ~80% and weight by ~50% while improving the system reliability and gathered data quality.
- Planned a test campaign that proved the root cause of an anomalous increment on opening pressures.
- Led, as Test Conductor, multiple actuation, fatigue, ultimate load, environmental, tensile/compression and cycle life tests.
- Analyzed and characterized the geometric degradation and opening pressure through different conditions.

SKILLS

Software: Python, MATLAB, CFD (ANSYS Fluent, OpenFoam), FEA (ANSYS), Siemens NX, SolidWorks, Inventor, CATIA V5, KiCad, LabVIEW, Arduino, GD&T.

Physical: Piping, Sensor and Instrumentation Set-up, Sensors Calibration, Manual Machining, TIG Welding, Cryogenics Handling, High-Pressure Systems, Soldering, XRF Analyzer, Damage Inspection.